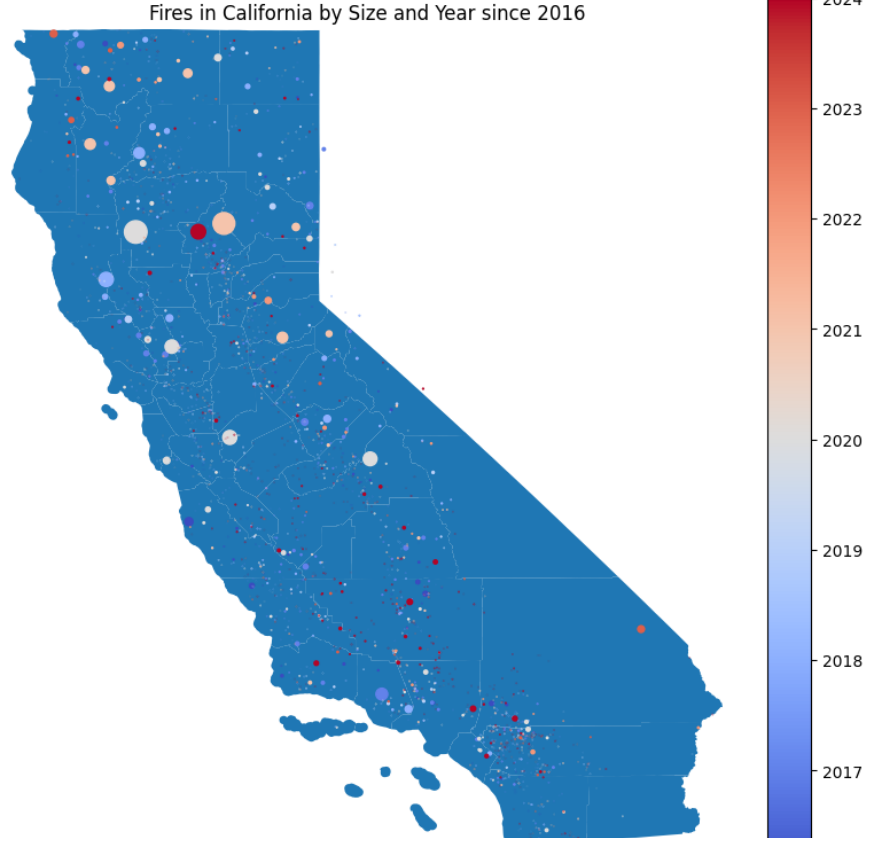


Introduction

Wildfires in California have become increasingly frequent and severe, presenting significant **risks to communities, ecosystems, and infrastructure** across the state. The increasing intensity and spread of these wildfires call for **advanced prediction tools** that can aid in the efficient allocation of resources, timely risk assessments, and **improved emergency response**. This project aims to develop a probability model to predict the **likelihood of wildfire occurrence** based on daily weather and fire-related data gathered from governmental sources. In addition, we aim to predict key resource-deploying metrics such as **fire cause and size**. By creating effective predictive models, this project intends to support fire departments and government agencies in deploying resources more strategically, potentially mitigating the impact of wildfires.



Data Background

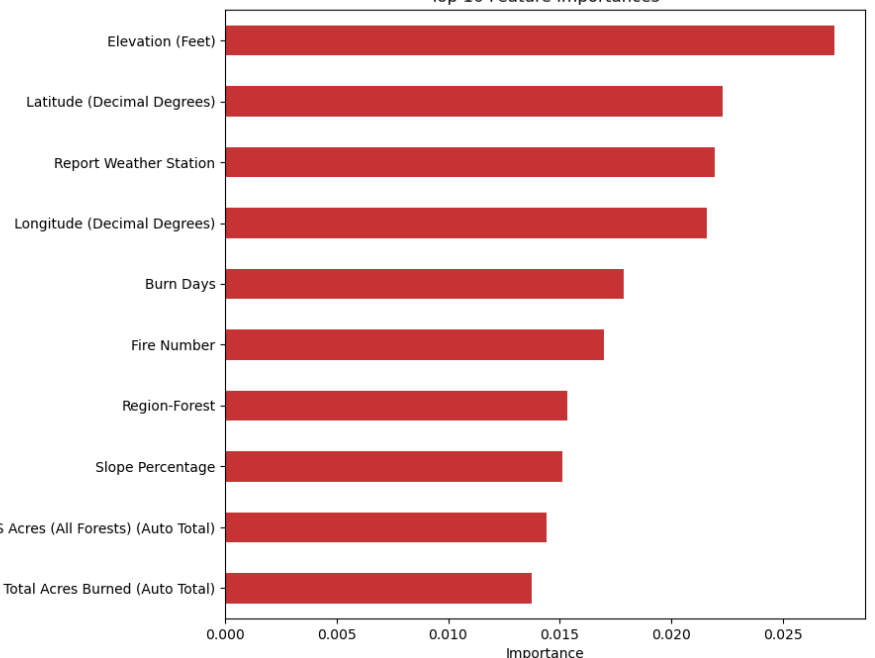
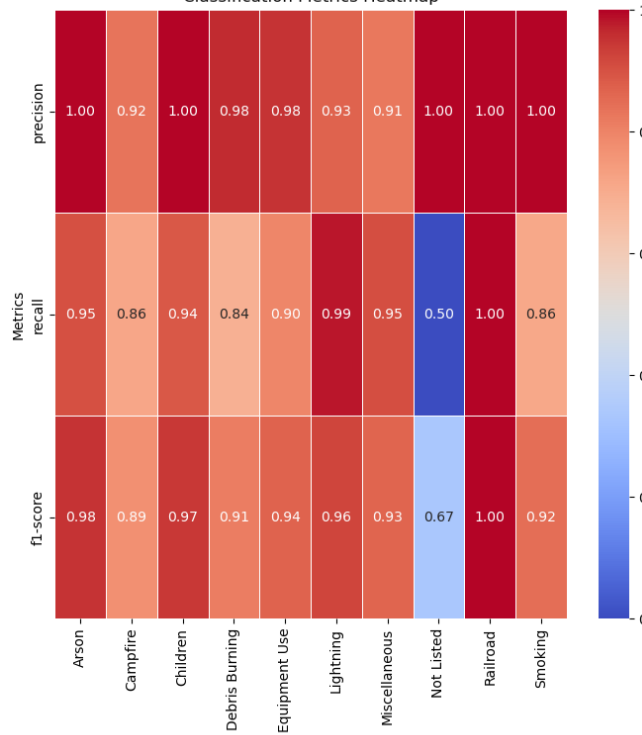
Our project integrates data from multiple sources to build robust datasets for wildfire prediction. We focus on three models: (1) Fire occurrence prediction, (2) Fire cause prediction, and (3) Fire size prediction. The datasets were compiled from:

- **California Department of Water Resources (CDEC):** Daily weather and precipitation data from 150 weather stations across California since 2016.
- **California Measurement Advisory Council (CALMAC):** Hourly wind and humidity data from 126 weather stations across California since 2016.
- **CALFIRE Incident Reports:** Historical records of wildfire occurrences, causes, and sizes.
- **FAMWeb Data Warehouse:** Comprehensive fire and fuel model data, which includes information on fire causes, total acres burned, and vegetation type.

We preprocessed and combined these datasets to ensure consistency across time, geography, and features. The first model predicts wildfire occurrence using historical weather and wind data. The second model leverages vegetation, location and temporal data to predict fire causes and sizes, which are critical for resource allocation and preparedness.

Fire Cause Prediction

We used Random Forest (RF), XGBoost, and Ensemble methods to predict wildfire causes, achieving accuracies of **93.85%**, **88%**, and **92.2%**, respectively. Hyperparameter tuning optimized the model's depth and number of estimators. Key predictors include **elevation, latitude, and weather station**—a location indicator.



Intuitively, it is likely that fire cause is closely related to the location of the incident, especially within California's diverse geography.

Fire Size Prediction: Insights and Performance

We approached fire size prediction as a **regression problem**, using a **log-transformation** on the target variable (total acres burned) to stabilize variance and improve model performance.

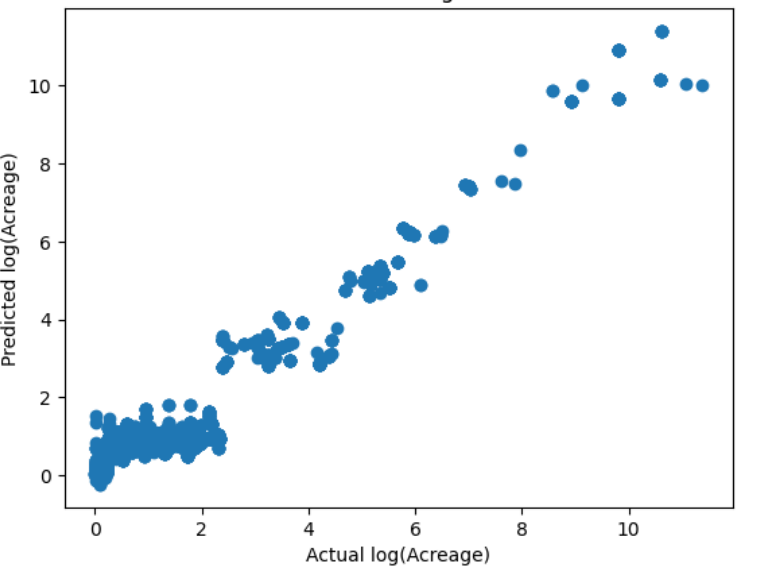
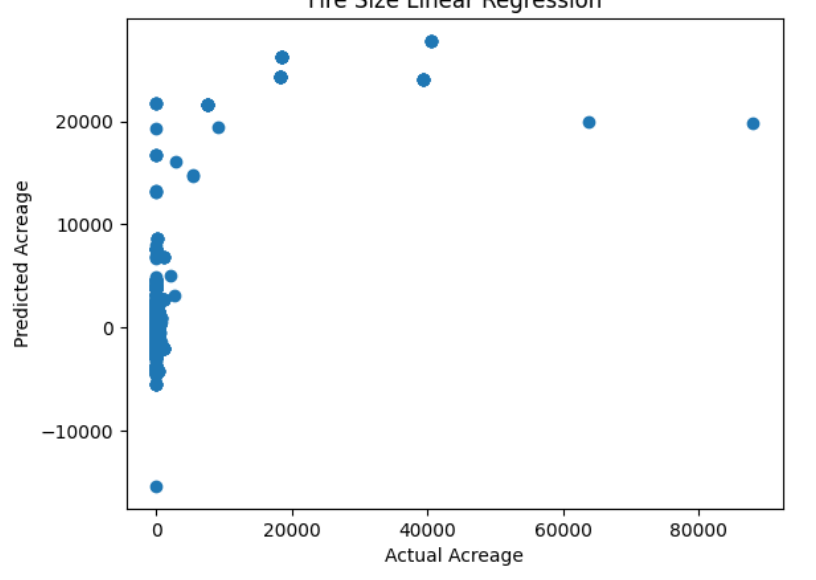


Figure 4. Comparison of predicted vs. actual fire sizes before and after log transformation.

The top-performing model was **XGBoost**, which minimized:

$$\mathcal{L}(f) = \sum_{i=1}^n \ell(y^{(i)}, f(x^{(i)})) + \gamma J + \frac{1}{2} \lambda \sum_{j=1}^J w_j^2,$$

where ℓ is mean squared error, J is the number of leaves, and w_j the prediction for leaf j . Hyperparameters included Learning rate (η), Maximum tree depth, and subsample ratio.

Model	RMSE	MAE	R ²
XGBoost	0.410	0.253	0.953
LightGBM	0.416	0.262	0.951
Gradient Boosting	0.426	0.266	0.949
Random Forest	0.459	0.263	0.941
Ridge	0.472	0.335	0.937

Table 3. Model Comparison for Fire Size Prediction (Log-Transformed)

Key Results

- **Fire Occurrence Prediction:** Achieved 74.7% accuracy on validation data.
- **Fire Cause Prediction:** Random Forest model outperformed others with a 93.35% accuracy.
- **Fire Size Prediction:** XGBoost achieved the best RMSE (0.41) and R^2 (0.95).

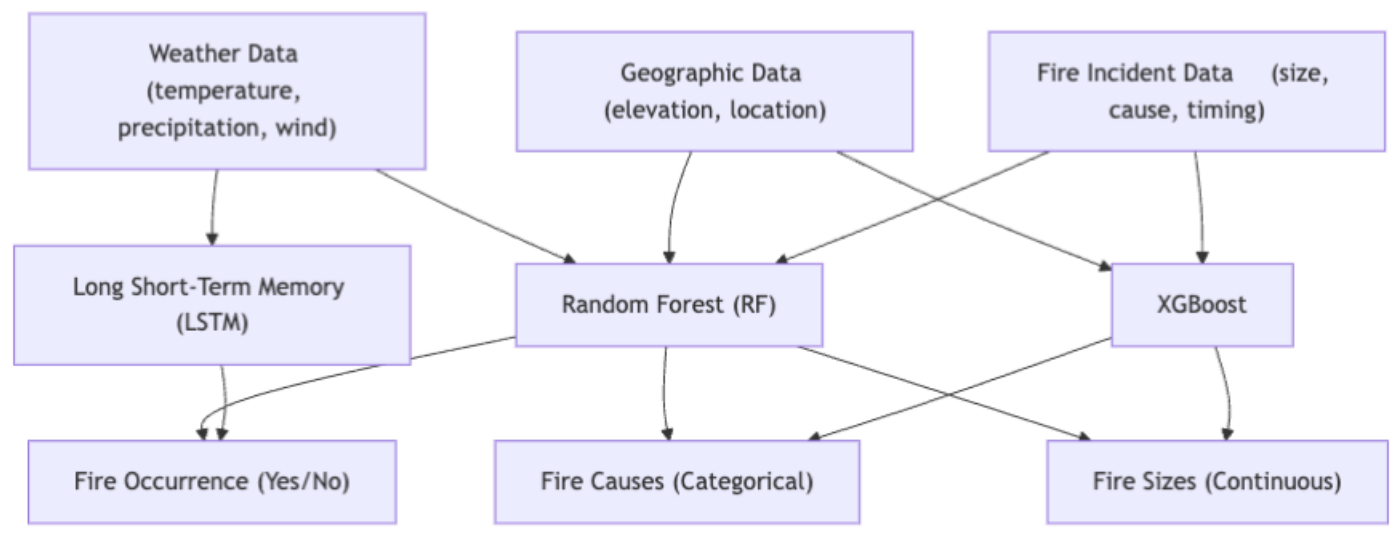
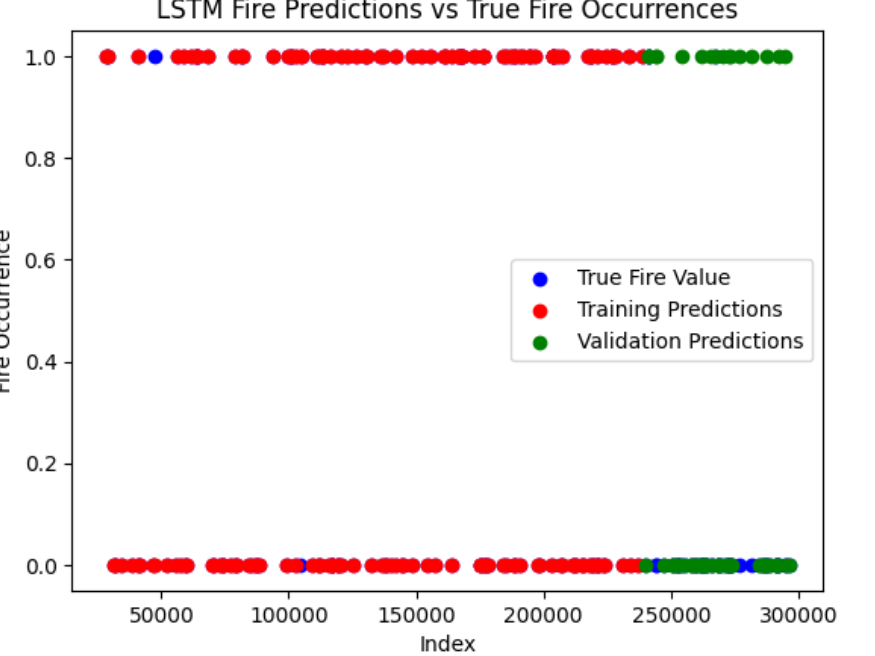
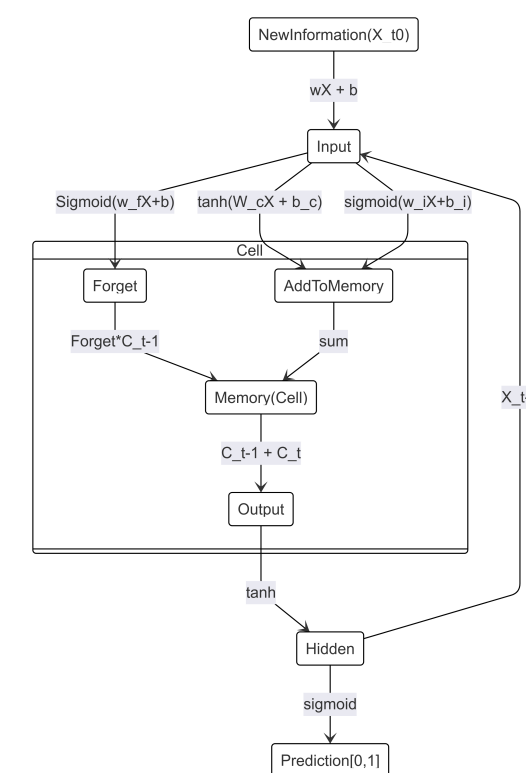


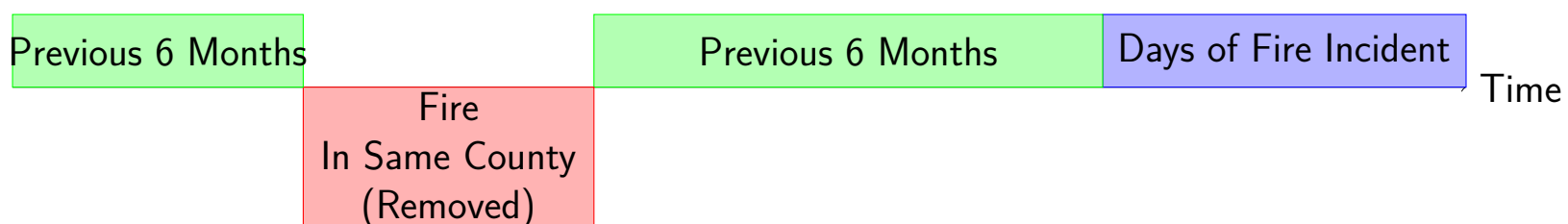
Figure 2. Flowchart of inputs, models, and outputs.

Time Series Classification: Fire Occurrence



(a) LSTM model (b) Occurrence Predictions

We used a *Long Short-Term Memory* (LSTM) model to predict whether a fire would occur based on weather data. We built the "look-back" in our data in the following way:



with a training accuracy of 97.9% and a validation accuracy of **74.7%**. We suspect the model is overfit due to this difference, but the results are promising. Likely causes of overfitting include data dropouts or sample biases.

	No Fire (Predicted)	Fire (Predicted)
No Fire (Actual)	30133	10917
Fire (Actual)	105	2398

Table 2. Confusion Matrix for LSTM